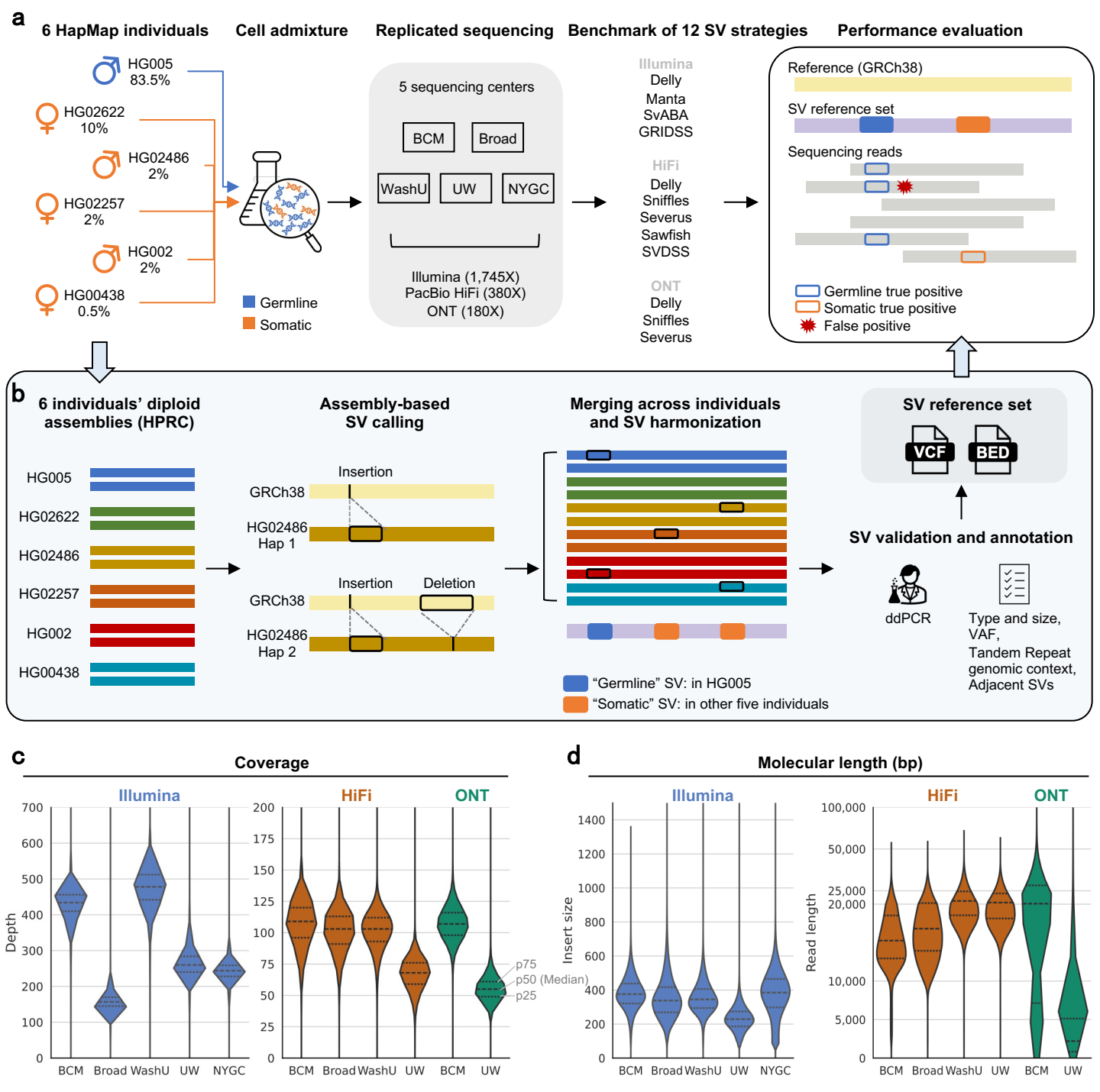

a
6 HapMap individuals
Cell admixture
Replicated sequencing
Benchmark of 12 SV strategies
Performance evaluation
HG005 83.5%
HG02622 10%
HG02486 2%
HG02257 2%
HG002 2%
HG00438 0.5%
Germline
Somatic
5 sequencing centers
BCM
Broad
WashU
UW
NYGC
Illumina (1,745X)
PacBio HiFi (380X)
ONT (180X)
Illumina
Delly
Manta
SvABA
GRIDSS
HiFi
Delly
Sniffles
Severus
Sawfish
SVDSS
ONT
Delly
Sniffles
Severus
Reference (GRCh38)
SV reference set
Sequencing reads
Germline true positive
Somatic true positive
False positive
b
6 individuals' diploid assemblies (HPRC)
Assembly-based SV calling
Merging across individuals and SV harmonization
SV reference set
VCF
BED
HG005
HG02622
HG02486
HG02257
HG002
HG00438
Insertion
GRCh38
HG02486 Hap 1
Insertion
Deletion
GRCh38
HG02486 Hap 2
SV validation and annotation
ddPCR
Type and size, VAF, Tandem Repeat genomic context, Adjacent SVs
"Germline" SV: in HG005
"Somatic" SV: in other five individuals
c
Coverage
Illumina
HiFi
ONT
Depth
BCM
Broad
WashU
UW
NYGC
p75
p50 (Median)
p25
d
Molecular length (bp)
Illumina
HiFi
ONT
Insert size
Read length
BCM
Broad
WashU
UW
NYGC